

STSCI 4780/5780

Continuous parameter estimation, cont'd

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Recap: Discrete and continuous spaces

We are calculating $P(H_i|\dots)$, $P(D_{\text{obs}}|\dots)$ over spaces of alternatives labeled by discrete or continuous parameters.

$P()$ is a real-valued function of (logical) *arguments*, e.g., $H_i|\mathcal{C}$.

Discrete spaces

Alternatives: $H_1, H_2 \dots$ for $i \in \mathbb{Z}$

$p_i \equiv P(H_i|\dots)$ is a *probability mass function* (PMF)

May use other similar symbols: $p(i)$, f_i , $g(i)$

Continuous spaces

Alternatives: H_θ for $\theta \in \mathbb{R}$

$p(\theta)d\theta \equiv P(\theta \in [\theta, \theta + d\theta]|\dots)$ defines a *probability density function* (PDF), $p(\theta)$

May use other similar symbols: $f(\theta)$, $g(\theta)$

More terminology

Functions, distributions, & measures

Mapping or **map**: An input/output relationship

PMF, PDF: Input is a label for a *single member* of the hypothesis/sample space (a set); output is a real-valued probability $\in [0, 1]$, or probability density (≥ 0):

$$p_i; \quad p(\theta)$$

Probability distribution or **measure**: Input is a *subset* in the space; output is a probability $\in [0, 1]$:

$$\mathcal{P}(S) = \sum_{i \in S} p_i; \quad \mathcal{P}(S) = \int_{\theta \in S} d\theta p(\theta)$$

Probability distribution function: Input is a label for a *single member defining a subset*; e.g., *cumulative distribution function* (**CDF**):

$$F_i = \sum_{j \leq i} p_j; \quad F(\theta) = \int_{\theta' < \theta} d\theta' p(\theta')$$

Recap of inference with Bernoulli/binomial data

Setup

$\mathcal{C}$ specifies existence of two outcomes, S and F , in each of N cases or trials; for each case or trial, the probability for S is α ; for F it is $(1 - \alpha)$

The trial probabilities are *IID* (independent and identically distributed)

H_i = Statements about α , the probability for success on the next trial $\rightarrow$ seek $p(\alpha|D, \mathcal{C})$

Adopt a *flat/uniform prior* as a default expression of initial ignorance about α — two motivations

Posterior (using sequence, binomial, negative binomial data)

$$p(\alpha|D, \mathcal{C}) = \frac{(N+1)!}{n!(N-n)!} \alpha^n (1-\alpha)^{N-n}$$

A *Beta distribution*.

Beta distribution (in general)

A two-parameter family of distributions for a quantity α in the unit interval $[0, 1]$:

$$p(\alpha|a, b) = \frac{1}{B(a, b)} \alpha^{a-1} (1-\alpha)^{b-1}$$

A PDF over possible 2-outcome PMFs

The beta-binomial conjugate model

Generalize from the flat prior to a $\text{Beta}(\alpha|a, b)$ prior for α

$$\begin{aligned} p(\alpha|n, M') &\propto \text{Beta}(\alpha|a, b) \times \text{Binom}(n|\alpha, N) \\ &\propto \alpha^{a-1}(1-\alpha)^{b-1} \times \alpha^n(1-\alpha)^{N-n} \\ &\propto \alpha^{n+a-1}(1-\alpha)^{N-n+b-1} \end{aligned}$$

$\Rightarrow$ the posterior is $\text{Beta}(\alpha|n+a, N-n+b)$

When the prior and likelihood are such that the posterior is in the same family as the prior, the prior and likelihood are said to comprise a *conjugate* pair

A Beta prior is a conjugate prior for the Bernoulli process, and for the binomial and negative binomial sampling distributions

Conjugacy $\rightarrow$ it's easy to chain inferences from multiple experiments

Probability & frequency

Recall $\hat{\alpha} = \frac{n}{N}$, the *relative frequency* of successes (uniform/flat prior); also $\sigma_{\alpha} \approx \frac{\sqrt{n}}{N}$ for $N, n \gg 1$

Frequencies arise when modeling repeated trials, or repeated sampling from a population or ensemble.

Finite-sample frequencies are observables

- When available, can be used to *infer* probabilities for next trial
- When unavailable, can be *predicted*

Bayesian/Frequentist relationships

- Relationships between probability and frequency
- Long-run performance of Bayesian procedures in IID settings (no accumulation of information)

Probability & frequency in IID settings

Frequency from probability

Bernoulli's (weak) *law of large numbers*: In repeated IID trials, given $P(\text{success} | \dots) = \alpha$, predict

$$\frac{n_{\text{success}}}{N_{\text{total}}} \rightarrow \alpha \quad \text{as} \quad N_{\text{total}} \rightarrow \infty$$

B. argued this justified estimating a next-trial probability with a (finite-sample) frequency—“*Bernoulli's swindle*”

Probability from frequency

Bayes's “An Essay Towards Solving a Problem in the Doctrine of Chances” → First use of Bayes's theorem

Compute *posterior probability* for success in next trial of IID sequence:

$$\mathbb{E}(\alpha) \rightarrow \frac{n_{\text{success}}}{N_{\text{total}}} \quad \text{as} \quad N_{\text{total}} \rightarrow \infty$$

If $P(\text{success} | \dots)$ does not change from sample to sample, it may be readily estimated using the observed relative frequency

Poisson process:

A continuous analog of the Bernoulli process

Bernoulli process and binomial distribution

Bernoulli process with success probability α produces binary sequences:

011001001100100110101001000100001...

Report n , the count of 1s in a sequence of length $N \rightarrow$
binomial distribution:

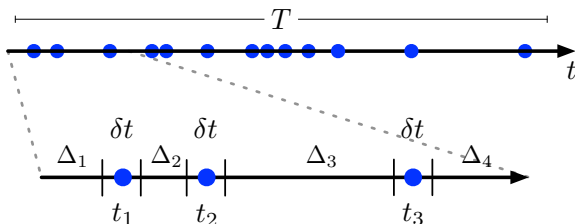
$$\begin{aligned}\mathcal{L}(\alpha) &\equiv p(n|\alpha, \mathcal{C}) \\ &= \frac{N!}{n!(N-n)!} \alpha^n (1-\alpha)^{N-n}\end{aligned}$$

Expected number of successes in N trials:

$$\mathbb{E}(n) = \alpha N$$

Poisson point process and Poisson (counting) distribution

Poisson point process with *intensity* λ (rate per unit interval):



Report n , the number of events in an interval of size $T \rightarrow$
Poisson distribution:

$$\begin{aligned}\mathcal{L}(\lambda) &\equiv p(n|\lambda, \mathcal{C}) \\ &= \frac{(\lambda T)^n}{n!} e^{-\lambda T}\end{aligned}$$

Expected number of counts in T :

$$\mathbb{E}(n) = \lambda T$$

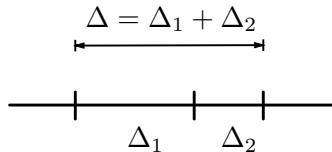
The Poisson distribution for counts from a point process

For occurrence/arrival of n events in an interval Δ , let's seek

$$f_n(\Delta) \equiv P(n \text{ events in } \Delta | \mathcal{P}),$$

where we'll figure out what we have to assume ($\mathcal{P}$) as we go

Partitioning an empty interval



$$f_0(\Delta) = P(\text{no events in } \Delta_1) \times P(\text{no events in } \Delta_2 | \text{no events in } \Delta_1) \parallel \mathcal{P}$$

As a simple modeling choice, let's *assume independence*:

$$P(\text{no events in } \Delta_2 | \text{no events in } \Delta_1) = P(\text{no events in } \Delta_2) \parallel \mathcal{P}$$

Independence implies

$$f_0(\Delta) = P(\text{no events in } \Delta_1) \times P(\text{no events in } \Delta_2) \quad || \mathcal{P}$$
$$\rightarrow \boxed{f_0(\Delta_1 + \Delta_2) = f_0(\Delta_1) \times f_0(\Delta_2)}$$

This is a **functional equation** for $f_0(\cdot)$; it has two solutions:

$$f_0(\Delta) = 0 \quad \text{so} \quad 0 = 0 \times 0$$

$$f_0(\Delta) = e^{-\lambda\Delta} \quad \text{so} \quad e^{-\lambda(\Delta_1+\Delta_2)} = e^{-\lambda\Delta_1} \times e^{-\lambda\Delta_2}$$

Let's use the *interesting one*:

$$\boxed{f_0(\Delta) = e^{-\lambda\Delta}}$$

Note this requires that *we specify a constant, λ*

Small interval behavior

What is the meaning of λ ? Note that

$$P(1 \text{ or more events in } \Delta | \mathcal{P}) = 1 - f_0(\Delta) = 1 - e^{-\lambda\Delta}$$

If Δ is small so that $\lambda\Delta \ll 1$, then $e^{-\lambda\Delta} = 1 - \lambda\Delta + O(\Delta^2)$

$$P(1 \text{ or more events in } \Delta | \mathcal{P}) = \lambda\Delta + O(\Delta^2)$$

The probability of seeing *at least* one event in a small interval is $\propto \Delta$ (and λ), and $\lambda \geq 0$

What about 2 events in a small interval?

$$\begin{aligned} P(2 \text{ or more events in } \Delta | \mathcal{P}) &= [1 - f_0(\Delta)] - f_1(\Delta) \\ &= \lambda\Delta - f_1(\Delta) + O(\Delta^2) \end{aligned}$$

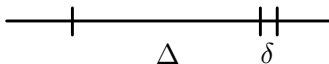
As another simplifying assumption, let's require that *events are simple*, so that this probability vanishes as $\Delta \rightarrow 0$ (multiple events can't happen at exactly the same instant/location):

$$f_1(\Delta) = \lambda\Delta + O(\Delta^2)$$

The probability of seeing *exactly* one event in a small interval is $\propto \Delta$, i.e., λ is a *rate parameter* (point process *intensity parameter*)

Extending an interval

To get a handle on the exact $f_n(\Delta)$ for $n > 0$, let's look at how $f_n(\Delta)$ changes if we grow the interval by a small amount δ :



Using the LTP we can write

$$f_n(\Delta + \delta) = f_n(\Delta)f_0(\delta) + f_{n-1}(\Delta)f_1(\delta) + f_{n-2}(\Delta)f_2(\delta) + \cdots$$

Let's exploit what we know about f_0 and small-interval behavior:

$$\begin{aligned} f_n(\Delta + \delta) &= f_n(\Delta)e^{-\lambda\delta} + f_{n-1}(\Delta)\lambda\delta + O(\delta^2) \\ &= f_n(\Delta)(1 - \lambda\delta) + f_{n-1}(\Delta)\lambda\delta + O(\delta^2) \\ f_n(\Delta + \delta) - f_n(\Delta) &= -\lambda\delta f_n(\Delta) + \lambda\delta f_{n-1}(\Delta) + O(\delta^2) \end{aligned}$$

Divide by δ and take $\lim_{\delta \rightarrow 0}$:

$$f'_n(\Delta) = -\lambda f_n(\Delta) + \lambda f_{n-1}(\Delta)$$

This is a recursive sequence of inhomogeneous differential equations—infininitely many!

$$f'_n(\Delta) = -\lambda f_n(\Delta) + \lambda f_{n-1}(\Delta)$$

Let's check $n = 0$, where there is no inhomogeneous term:

$$f'_0(\Delta) = -\lambda f_0(\Delta)$$

The solution is $Ce^{-\lambda\Delta}$, but since we know $f_0(0) = 1$, we know $C = 1$

For $n = 1$,

$$f'_1(\Delta) = -\lambda f_1(\Delta) + \lambda e^{-\lambda\Delta}$$

As an inspired guess (or using *variation of parameters*), try

$$f_1(\Delta) = \lambda\Delta e^{-\lambda\Delta}$$

$$\rightarrow f'_1(\Delta) = -\lambda^2\Delta e^{-\lambda\Delta} + \lambda e^{-\lambda\Delta}$$

which satisfies the differential eq'n

Iterating, we find:

$$f_n(\Delta) = \frac{(\lambda\Delta)^n}{n!} e^{-\lambda\Delta}$$

The Poisson distribution

If we model events distributed in an interval Δ such that:

- A single parameter, λ , governs the process
- With λ specified, probabilities for event counts in non-overlapping intervals are independent
- The events are simple (no two are at the same time/location)

then denoting these assumptions by $\mathcal{P} = \lambda, \mathcal{C}$ (and including the interval size in $\mathcal{C}$)

$$p(n|\lambda, \mathcal{C}) = \frac{(\lambda\Delta)^n}{n!} e^{-\lambda\Delta}$$

with λ corresponding to the event rate

We can show:

$$\mathbb{E}(n) = \lambda\Delta, \quad \text{Var}(n) = \lambda\Delta$$

“ $\lambda, \mathcal{C}$ ” is analogous to “ α, N IID trials” for binomial

Infer a Poisson rate from counts

Problem:

Observe n counts in T ; infer rate (intensity), r

Likelihood

Poisson distribution:

$$\begin{aligned}\mathcal{L}(r) &\equiv p(n|r, \mathcal{C}) \\ &= \frac{(rT)^n}{n!} e^{-rT}\end{aligned}$$

Prior

Two simple “uninformative” standard choices:

- r known to be *nonzero*: it is a scale parameter; scale invariance $\rightarrow$

$$p(r|\mathcal{C}) = \frac{1}{\ln(r_u/r_l)} \frac{1}{r}$$

This corresponds to a flat prior on $\lambda = \log r$

- r may *vanish*; require prior predictive $p(n|\mathcal{C}) \sim \text{Const}$:

$$p(r|\mathcal{C}) = \frac{1}{r_u}$$

The *reference prior* (“uninformative” in an asymptotic, information-theoretic sense) is $p(r|\mathcal{C}) \propto 1/r^{1/2}$

Prior predictive

Adopting a flat (uniform) prior,

$$\begin{aligned} p(n|\mathcal{C}) &= \frac{1}{r_u} \frac{1}{n!} \int_0^{r_u} dr (rT)^n e^{-rT} \\ &= \frac{1}{r_u T} \frac{1}{n!} \int_0^{r_u T} d(rT) (rT)^n e^{-rT} \\ &\approx \frac{1}{r_u T} \quad \text{for } r_u \gg \frac{n}{T} \end{aligned}$$

Posterior

A *gamma distribution*:

$$p(r|n, \mathcal{C}) = \frac{T(rT)^n}{n!} e^{-rT}$$

Gamma Distributions

A 2-parameter family of distributions over nonnegative x , with shape parameter α and scale parameter λ (or inverse scale ϵ):

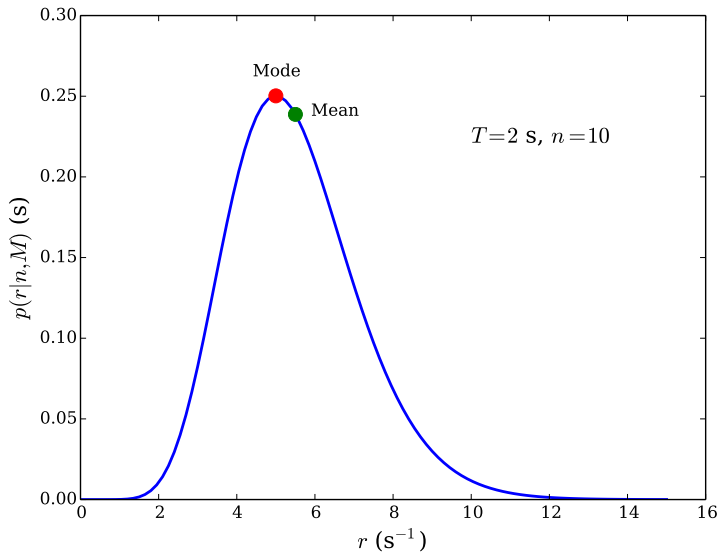
$$\begin{aligned} p_{\Gamma}(x|\alpha, \lambda) &\equiv \frac{1}{\lambda \Gamma(\alpha)} \left(\frac{x}{\lambda}\right)^{\alpha-1} e^{-x/\lambda} \\ &\equiv \frac{\epsilon}{\Gamma(\alpha)} (x\epsilon)^{\alpha-1} e^{-x\epsilon} \end{aligned}$$

Moments:

$$\mathbb{E}(x) = \alpha\lambda = \frac{\alpha}{\epsilon} \qquad \text{Var}(x) = \lambda^2\alpha = \frac{\alpha}{\epsilon^2}$$

Our posterior corresponds to $\alpha = n + 1$, $\lambda = 1/T$.

- Mode $\hat{r} = \frac{n}{T}$; mean $\langle r \rangle = \frac{n+1}{T}$ (shift down 1 with $1/r$ prior)
- Std. dev'n $\sigma_r = \frac{\sqrt{n+1}}{T}$; credible regions found by integrating (can use incomplete gamma function)



Conjugate prior

Note that a gamma distribution prior is the conjugate prior for the Poisson sampling distribution:

$$\begin{aligned} p(r|n, M') &\propto \text{Gamma}(r|\alpha, \epsilon) \times \text{Pois}(n|rT) \\ &\propto r^{\alpha-1} e^{-r\epsilon} \times r^n e^{-rT} \\ &\propto r^{\alpha+n-1} \exp[-r(T + \epsilon)] \end{aligned}$$

Useful conventions

- Use a flat prior for a rate that may be zero
- Use a log-flat prior ($\propto 1/r$) for a nonzero scale parameter
- Use proper (normalized, bounded) priors
- Plot posterior with abscissa that makes prior flat (use $\log r$ abscissa for scale parameter case)

Supplementary material

Binomial for rare events

How many Cornell students share your birthday?

- $N \approx 24,000 \gg 1$
- $\alpha \approx \frac{1}{365} \ll 1$
- Expected number $\mu \equiv \mathbb{E}(n) = \alpha N \approx 66 \gg 1$

1000 bacteria are mixed in a liter of water. How many are in a 0.1 ml sample?

- $N = 1000 \gg 1$
- $\alpha = 10^{-4}$
- Expected number $\mu \equiv \mathbb{E}(n) = \alpha N = 0.1 \ll 1$

Seek an approximation for $p(n|\dots)$ for small α , but not necessarily small μ (or n): A rare event can happen many times in a very large sample

Recall the binomial sampling distribution for n successes in N trials, given success probability α :

$$p(n|\alpha, N) = \frac{N!}{n!(N-n)!} \alpha^n (1-\alpha)^{N-n}$$

Expected number of successes $\mu \equiv \mathbb{E}(n) = \alpha N$

Recursion relation:

$$\begin{aligned} \frac{p(n)}{p(n-1)} &= \frac{N!}{n!(N-n)!} \frac{(n-1)!(N-n+1)!}{N!} \frac{\alpha}{1-\alpha} \\ &= \frac{N-n+1}{n} \frac{\alpha}{1-\alpha} \end{aligned}$$

Consider the limit where $N \rightarrow \infty$ and $\alpha \rightarrow 0$, but with $\mu = \alpha N$ fixed and not necessarily small (but $\mu \ll N$); focus on $n \sim \mu$ so $n \ll N$ as well:

$$\frac{p(n)}{p(n-1)} \approx \frac{N\alpha}{n} = \frac{\mu}{n}$$

In that same limit, writing α in terms of μ and N ,

$$p(0) = (1 - \alpha)^N = \left(1 - \frac{\mu}{N}\right)^N \approx e^{-\mu}$$

Now evaluate $p(n)$ using the recurrence relation:

$$p(1) = \frac{\mu}{1} \times p(0) = \mu e^{-\mu}$$

$$p(2) = \frac{\mu}{2} \times p(1) = \frac{\mu^2}{2} e^{-\mu}$$

$$p(n) = \frac{\mu}{n} \times p(n-1) = \frac{\mu^n}{n!} e^{-\mu} = \text{Poisson}$$

The *Poisson limit theorem* or *law of rare events* (events rare in *proportion*, though possibly numerous)